Mapping and Review of Camera Models for Polarization Navigation

Project Progress Report

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Outline

Motivation

Currently there are only two **self-sufficient** means for computation of bearing that has been widely accepted in technical systems [1]:

- **Magnetic compass**:
  - Suffers from distortion of the magnetic field by electric currents in the airframe
  - Geomagnetic field is not uniform which requires additional compensation scheme
- **Gyrocompass**:
  - Have strong anti-interference ability but accumulate error over time [2]
  - Relative position according to the initial position instead of absolute position
  - Long time to stabilize in case of a power failure

[1] A. Dawadee, Towards an optical aircraft navigation system, Applied Mechanics and Materials 629: 321-326 (2014).
[2] J. Liu et al, Global positioning method based on polarized light compass system, Review of scientific instruments 89: 054503 (2018).

A solution inspired by nature...

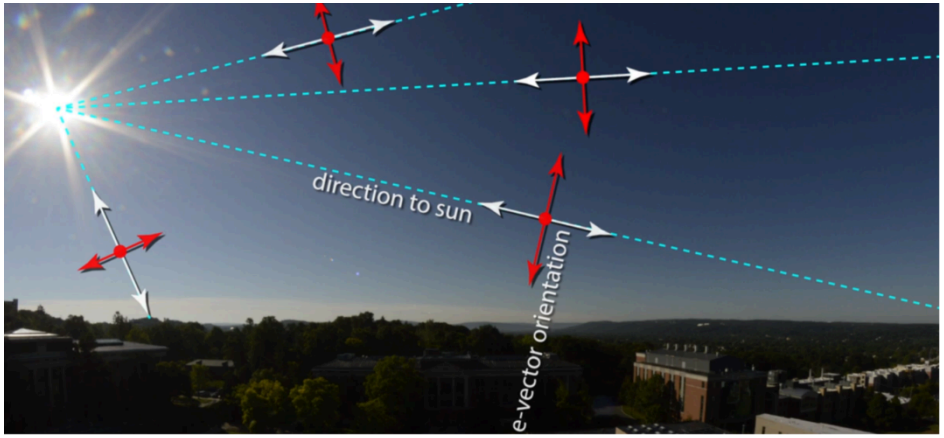
Case study: The Sahara desert ant

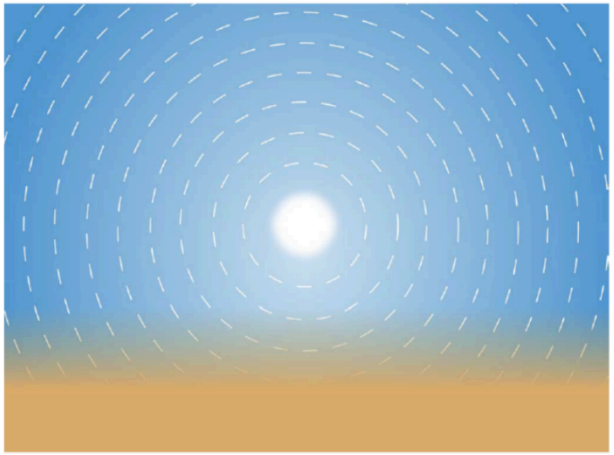
- Desert environment poses existential challenges:
  - Pheromone trails employed by other ant species for navigation cannot survive high temperature
  - No visual cues for use as path markers
  - Foraging durations are limited by intense heat
- Despite challenges, the ant manages to thrive!
- The remarkable navigation ability is achieved with the limited resources available

What is the underlying physical principle?

Polarization of Skylight

- When sunlight passes through the atmosphere, it becomes partially polarized
- Electric field for the polarized light becomes tangent to the sun's circumference
- Polarization becomes stronger with increasing distance from the sun
- The position of sun can be determined from analyzing the polarization of light





Camera Models

Explain:

- What are camera models?
- Why are they useful for polarization navigation?

Research Questions

RQ 0

This is my mapping research question.

RQ 1

RQ 1

Search Method: Search String Piloting

This is the search string

1. Test results for control group of known good publications

Search Method: Publication Collection

1. Download results from databases as .bib files
   - Scopus, IEEExplore, Web of Science, ACM
   - Title, author, abstract, venue, date, etc.
2. Combine results and deduplicate
   - Deduplication by unique DOI
3. Assign unique citation key to each entry

Search Method: Application of Exclusion Criteria

1. English language title and abstract are not available
2. DOI is not available (or does not exist)
3. No contribution toward application of polarization to navigation
4. No application of a skylight polarization model in methodology

Search Results